实操手册

登录线上实验室环境

官方申请地址：https://flagos.io/login

官方文档：https://docs.flagos.io/projects/onlinelaboratory/zh-cn/latest/onlinelaboratory.html

查询GPU：

代码块

```
npu-smi info
```

```
root@6edf64f2602a:~# npu-smi info
+------------------------------------------------------------------------------------------------+
| npu-smi 25.5.0                   Version: 25.5.0                                               |
+---------------------------+---------------+-----------------------------------------------------+
| NPU   Name                | Health        | Power(W)    Temp(C)           Hugepages-Usage(page)|
| Chip  Phy-ID              | Bus-Id        | AICore(%)   Memory-Usage(MB)  HBM-Usage(MB)        |
+===========================+===============+=====================================================+
| 7     Ascend910           | OK            | 165.4       41                0    / 0            |
| 0     14                  | 0000:81:00.0  | 0           0    / 0          60030/ 65536         |
+---------------------------+---------------+-----------------------------------------------------+
| 7     Ascend910           | OK            | -           42                0    / 0            |
| 1     15                  | 0000:83:00.0  | 0           0    / 0          2881 / 65536         |
+===========================+===============+=====================================================+
+---------------------------+---------------+-----------------------------------------------------+
| NPU     Chip              | Process id    | Process name             | Process memory(MB)      |
+===========================+===============+=====================================================+
| 7       0                 | 72637         | VLLMEngineCor            | 56880                   |
| 7       0                 | 72090         | vllm                     | 70                      |
+===========================+===============+=====================================================+
```

安装Qwen3-4B模型

1. 安装ModelScope

代码块

```
pip install modelscope
```

2. 通过 `modelscope` 下载Qwen官方开源模型Qwen3-4B的模型权重：

代码块

```
modelscope download --model Qwen/Qwen3-4B  --local_dir ./Qwen3-4B
```

安装FlagGems、FlagTree、vllm-plugin-FL

1. 安装FlagGems

代码块

```
pip install -U scikit-build-core==0.11 pybind11 ninja cmake
git clone https://github.com/flagos-ai/FlagGems
cd FlagGems
pip install --no-build-isolation -e .
```

测试安装成功:

代码块

```
pip show flag_gems
```

```
Name: flag_gems
Version: 4.2.1rc0
Summary: FlagGems is a function library written in Triton.
Home-page: https://github.com/FlagOpen/FlagGems
Author:
Author-email: Zhixin Li <strongspoon@outlook.com>, Tongxin Bai <waffle.bai@gmail.com>, Yuming Huang <jokmingwong@gmail.com>, Feiyu Chen <iclementine@outlook.com>
License: Apache Software License
Location: /usr/local/python3.11.13/lib/python3.11/site-packages
Editable project location: /root/FlagGems
Requires: packaging, pybind11, PyYAML, sqlalchemy, torch
Required-by:
```

2. 安装FlagTree

代码块

```
RES="--index-url=https://resource.flagos.net/repository/flagos-pypi-hosted/simple --trusted-host=https://resource.flagos.net"
python3 -m pip install flagtree==0.4.0+ascend3.2 $RES
```

测试安装成功:

代码块

```
pip show flagtree
```

```
Summary: A unified compiler supporting multiple AI chip backends for custom Deep Learning operations, which is forl
ed from triton-lang/triton.
Home-page: https://github.com/triton-lang/triton/
Author: FlagOS
Author-email: contact@flagos.io
License:
Location: /usr/local/python3.11.13/lib/python3.11/site-packages
Requires:
Required-by:
```

3. 安装vllm-plugin-FL

代码块

```
git clone https://github.com/flagos-ai/vllm-plugin-FL
cd vllm-plugin-FL
pip install -r requirements.txt
pip install --no-build-isolation -e .
```

测试软件版本：

代码块

```
cd vllm-plugin-FL
git log
```

```
root@6edf64f2602a:~/vllm-plugin-FL# git log
commit ba008211c1c9646e19290e832a7f7775f7d2944f (HEAD -> main, origin/main
, origin/HEAD)
Author: xin2an <yuxinan@perfxlab.com>
Date:   Sat Feb 28 22:02:09 2026 +0800

    The Ascend platform adds a new Flagos blacklist operator. (#63)

    ### PR Category
    <!-- One of [Core | Vendor | OP | Tools | Others ] -->
```

配置推理服务

1. 启动服务: 需要修改模型路径，例如：/root/Qwen3-4B

代码块

```
export VLLM_PLUGINS=fl
export TRITON_ALL_BLOCKS_PARALLEL=1
vllm serve --model /root/Qwen3-4B --served-model-name qwen --enforce-eager --
port 30000
```

注意: 启动推理服务时会自动启动FlagGems算子, 通过以下命令可以查询被启动的FlagGems算子:

代码块

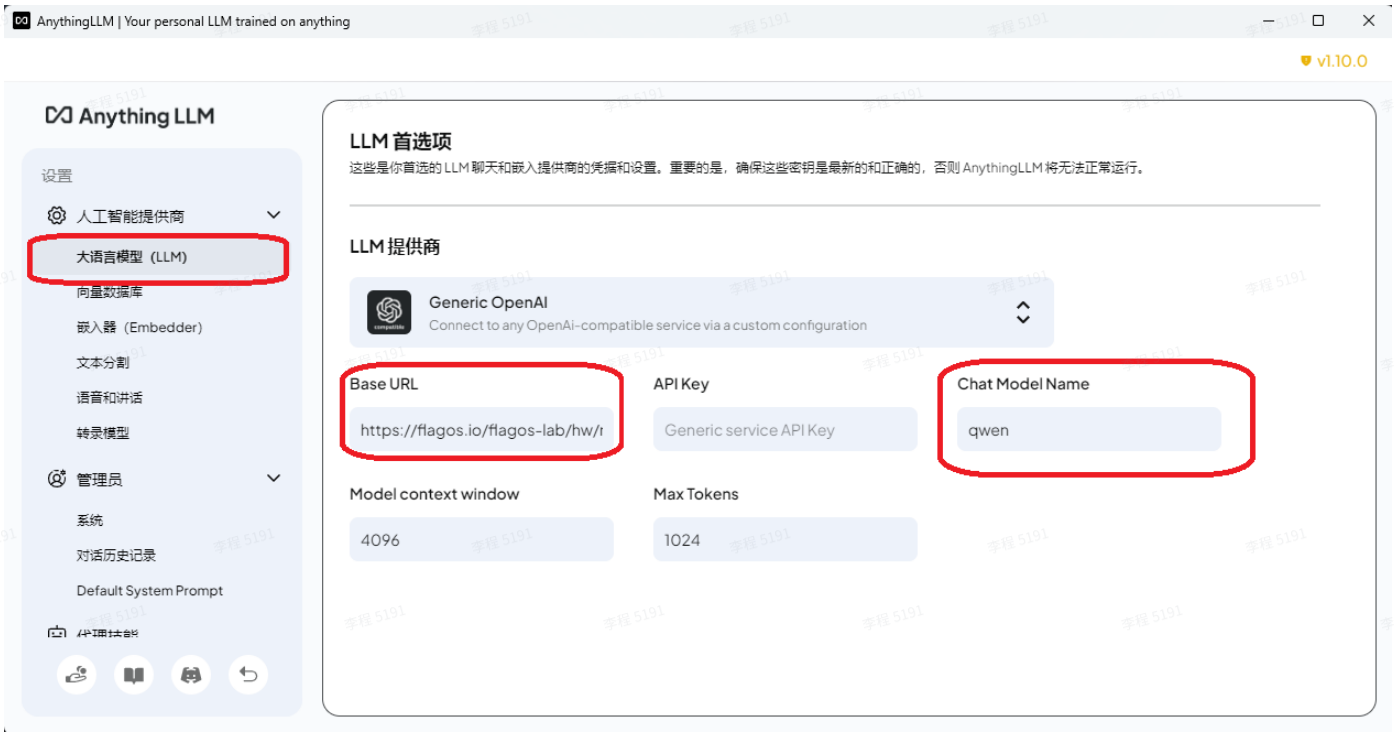
```
cat /tmp/flaggems_enable_oplist.txt
```

实验室外调用推理服务

如果您在启动服务时，配置了端口30000, 您可以在实验室外使用客户端连接实验室的服务器来创建机器人进行对话。步骤如下：

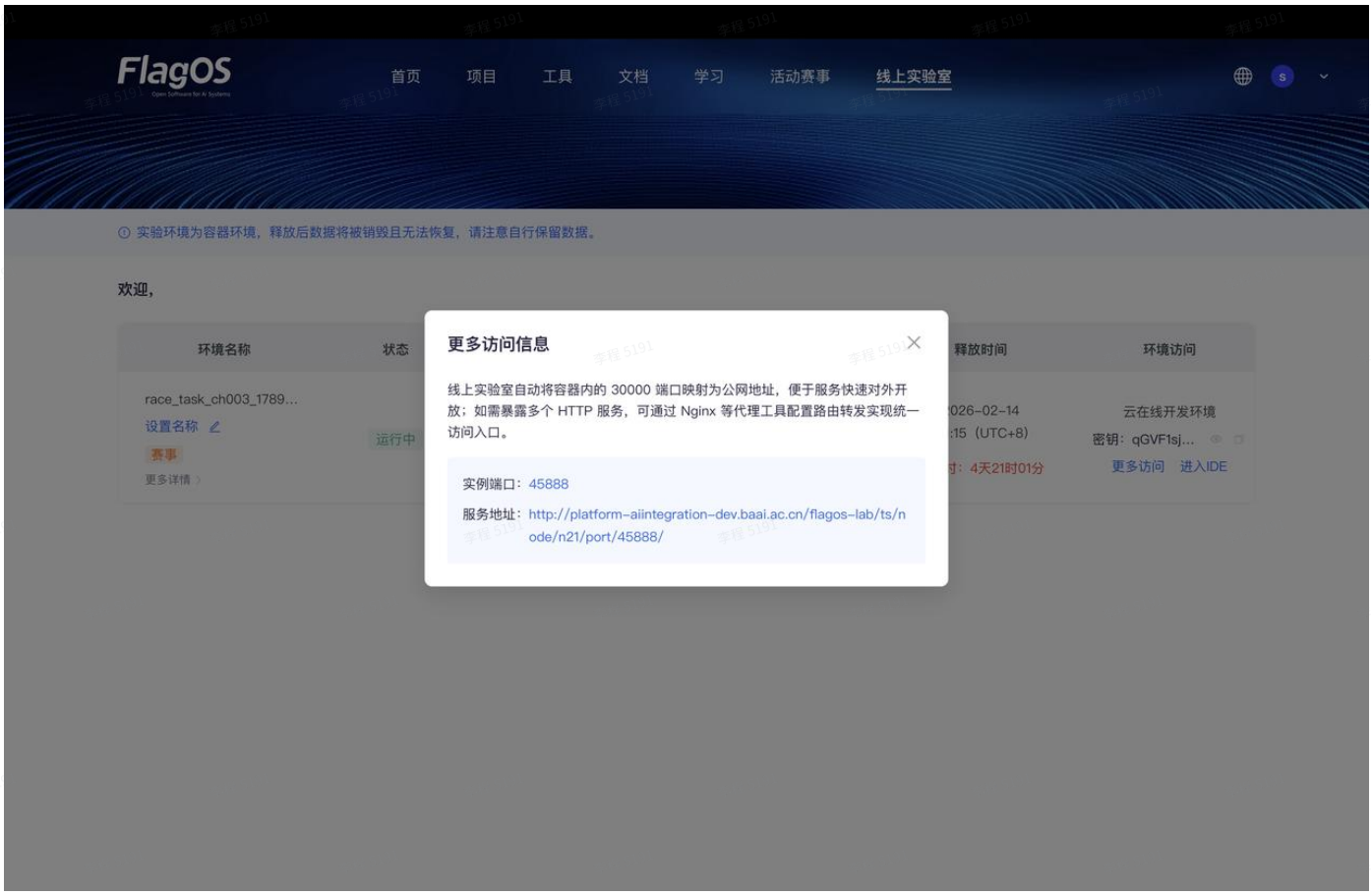
1. 下载AnythingLLM。
2. 点击小扳手按钮:

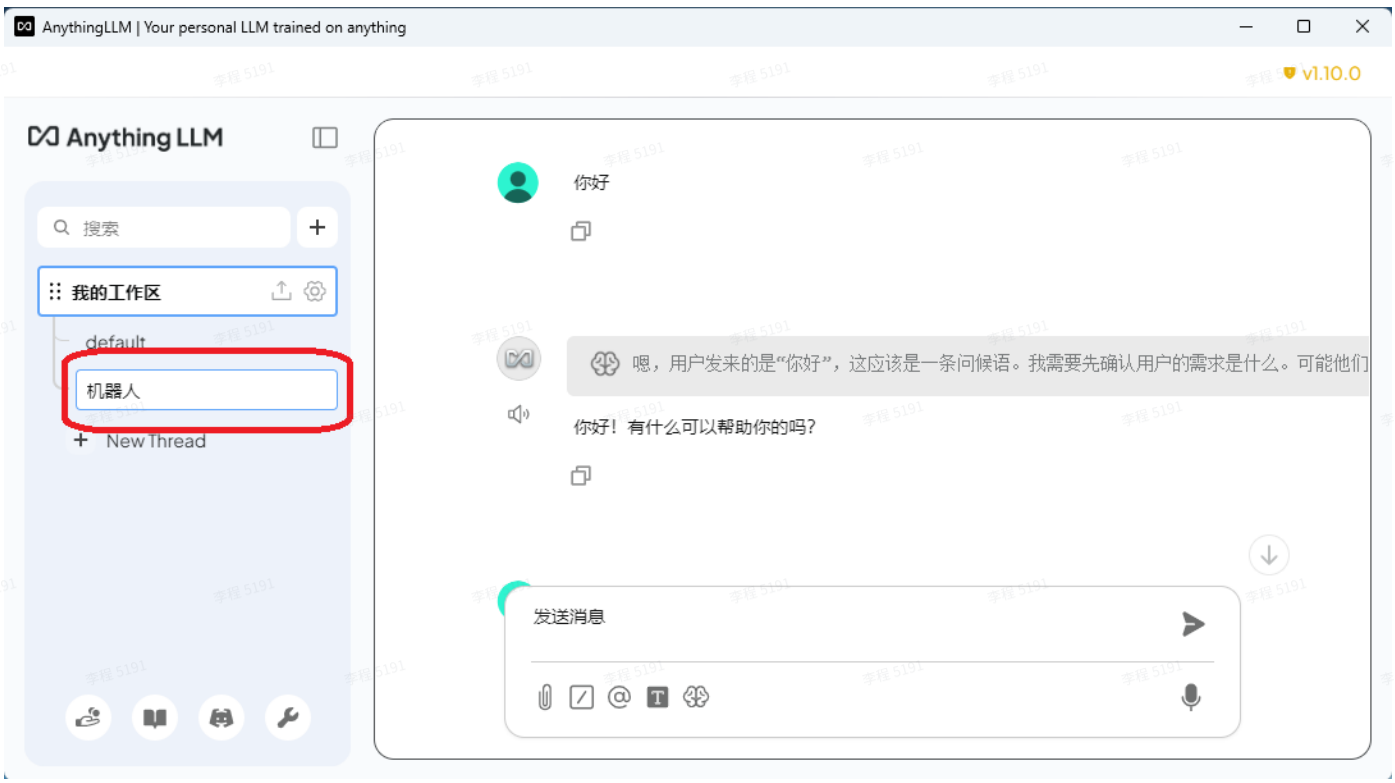
3. 选择**大语言模型（LLM）**并配置必要项：**Base URL**和**Chat Model Name**。不需要配置**API Key**。其它为可选项。



- **Base URL**：这里填写您申请线上实验室时的**服务地址**。您需要回到线上实验室，点击**更多访问**:

在弹出的对话框里，复制**服务地址**，并在服务地址后加**v1**，形成一个完整的IP地址。粘贴到AnthingLLM的**Base URL**里。



- **Chat Model Name**：这里填写您在启动服务时使用的命令 `vllm serve --model /root/Qwen3-4B --served-model-name qwen --enforce-eager --port 30000` 中的名字**qwen**。

4. 点击小扳手回到**我的工作区**, 在**我的工作区**点击**New Thread**, 命名为“机器人”，在消息窗口进行对话。